

Assignment 1 Report

All results are generated by using Python with pandas, numpy and statsmodels.api imported.

The data used are “tickers.csv”, “mktcap.csv”, “univ_h.csv”, and “adjusted.csv”.

Q1:

The data processing steps are:

- 1) Generate codes of GISC sectors and GISC industries for all tickers in the file “tickers.csv”.
- 2) Obtain the market cap data for each ticker at the beginning of 2008 and 2025, namely 2008/01/02 and 2025/01/02.
- 3) Merge the market cap data with the tickers data on ticker identifiers with the ticker data as the primary table, yielding “df_merged”.
- 4) Generate those tickers that are included in the S&P 500 universe at the year of 2008 and 2025 respectively by using file “univ_h.csv”.
- 5) Filter the “df_merged” using the tickers that are included in the S&P 500 universe at the year of 2008 and then drop those data with missing GISC code to get “ticker2008”.
- 6) Calculate the size and market cap for each GISC sectors at the year of 2008. The code is `sec2008 = ticker2008.groupby('GISC sectors').agg(number=('GISC sectors','size'),mktcap=('mkt_cap_2008','sum'))`. Therefore, we include those without market cap data when calculating the size but exclude them when calculating the total market cap. This is because the tickers have too much missing market cap data so that we cannot fill them with data from other days.
- 7) Sort the data by “mktcap”.
- 8) Repeat the (5)(6)(7) steps and obtain the total market cap of each sector and industry at the beginning of year 2008 and year 2025.

The results are as follows:

The top three sectors at the beginning of 2008 are 40 (number of stocks: 70, market cap: 1.91e+06), 45 (number of stocks: 48, market cap: 1.737e+06), 10 (number of stocks: 29, market cap: 1.573e+06).

	number	mktcap
GISC sectors		
40	70	1.910081e+06
45	48	1.737036e+06
10	29	1.572799e+06
35	44	1.458662e+06
20	53	1.436799e+06
30	34	1.337934e+06
50	23	1.011713e+06
25	54	6.498488e+05
55	25	4.061667e+05
15	24	4.031101e+05
60	14	1.459626e+05

The top three industries at the beginning of 2008 are 101020 (number of stocks: 19, market cap: 1.259e+06), 401010 (number of stocks: 18, market cap: 7.507e+05), 201010 (number of stocks: 12, market cap: 7.130e+05).

		number	mktcap					
GISC industries								
101020	19	1.259032e+06						
401010	18	7.506683e+05	452010	5	2.201652e+05	601080	3	4.432347e+04
201010	12	7.130312e+05	201060	13	2.015221e+05	601070	4	4.295959e+04
352020	9	6.143383e+05	253010	10	1.872472e+05	201020	3	3.927646e+04
403010	23	5.389853e+05	255040	14	1.777939e+05	202010	4	3.766830e+04
451030	10	5.312700e+05	302030	2	1.771785e+05	151030	6	3.580474e+04
402030	19	4.407964e+05	302020	11	1.766712e+05	252010	8	3.148728e+04
501010	5	3.881952e+05	451020	5	1.689559e+05	303020	2	2.521859e+04
453010	16	3.879756e+05	502010	9	1.382640e+05	251020	2	2.494046e+04
452020	8	3.687178e+05	352010	4	1.271412e+05	253020	3	2.487787e+04
301010	9	3.590416e+05	151040	5	1.250726e+05	601040	2	2.441478e+04
351020	15	3.282866e+05	551030	11	1.205417e+05	601060	3	2.031755e+04
351010	11	3.140222e+05	255030	7	1.198314e+05	551050	1	1.412802e+04
101010	10	3.137676e+05	203010	4	1.174145e+05	201030	1	1.276853e+04
303010	4	3.019615e+05	201050	2	1.037227e+05	252020	3	1.195863e+04
302010	6	2.978623e+05	402020	4	9.493805e+04	551020	1	9.502719e+03
551010	12	2.619942e+05	402010	6	8.469252e+04	601030	1	9.027003e+03
502030	4	2.567324e+05	352030	5	7.487379e+04	203020	1	8.661649e+03
151010	12	2.349277e+05	203040	4	7.300779e+04	255010	1	7.562230e+03
502020	5	2.285211e+05	201040	3	6.703737e+04	151020	1	7.305034e+03
			452030	4	5.995102e+04	201070	1	6.776172e+03
			252030	5	5.848705e+04	251010	1	5.662751e+03
			202020	5	5.591224e+04	602010	1	4.920258e+03

The top three sectors at the beginning of 2025 are 45 (number of stocks: 69, market cap: 1.647e+07), 50 (number of stocks: 22, market cap: 7.777e+06), 40 (number of stocks: 73, market cap: 7.444e+06).

	number		mktcap
GISC sectors			
	45	69	1.646957e+07
	50	22	7.777047e+06
	40	73	7.443647e+06
	25	50	6.072731e+06
	35	61	5.227583e+06
	20	78	4.187276e+06
	30	38	3.201579e+06
	10	22	1.658627e+06
	55	31	1.171475e+06
	60	31	1.064923e+06
	15	28	9.539625e+05

The top three industries at the beginning of 2025 are 502030 (number of stocks: 4, market cap: 6.173e+06), 453010 (number of stocks: 19, market cap: 5.815e+06), 451030 (number of stocks: 21, market cap: 5.422e+06).

	number	mktcap						
GISC industries								
502030	4	6.173448e+06						
453010	19	5.815256e+06	502020	6	7.063237e+05	203010	4	1.993167e+05
451030	21	5.422167e+06	302010	7	6.361374e+05	252010	6	1.736911e+05
452020	8	3.907135e+06	151010	16	6.359026e+05	151040	4	1.420353e+05
402010	10	2.486720e+06	451020	8	5.719238e+05	601070	5	1.409100e+05
255030	2	2.345443e+06	303010	5	5.539054e+05	601060	7	1.387685e+05
402030	23	1.808837e+06	352030	10	5.451286e+05	201070	3	1.367377e+05
401010	13	1.729980e+06	452010	5	4.802212e+05	601050	4	1.331077e+05
352020	7	1.690411e+06	601080	10	4.606732e+05	101010	3	1.197123e+05
101020	19	1.538915e+06	203040	6	4.417970e+05	151030	6	1.110692e+05
301010	8	1.312754e+06	201040	7	3.990504e+05	601025	1	9.899800e+04
251020	3	1.312350e+06	302020	14	3.541414e+05	203020	3	8.950767e+04
351010	20	1.140119e+06	402020	4	3.475757e+05	602010	2	6.878917e+04
403010	23	1.070534e+06	501010	2	3.330809e+05	303020	2	6.735643e+04
351020	16	1.046448e+06	202020	9	3.226179e+05	151020	2	6.495533e+04
253010	18	1.044506e+06	202010	6	3.180251e+05	551050	2	6.019716e+04
201010	12	9.413027e+05	551030	10	3.100011e+05	201030	1	4.657146e+04
255040	10	9.138753e+05	502010	9	3.094326e+05	255010	3	3.822507e+04
352010	8	8.054763e+05	302030	2	2.772850e+05	551040	1	2.413956e+04
201060	17	8.030984e+05	452030	8	2.728699e+05	551020	1	2.149329e+04
551010	17	7.556438e+05	201020	8	2.719775e+05	251010	2	2.102234e+04
			501020	1	2.547616e+05	601030	1	1.202330e+04
			201050	2	2.172730e+05	601040	1	1.165276e+04
			252030	5	2.157465e+05	252020	1	7.872065e+03

Q2:

We first get the code of GISC industries for MCD, which is 253010 and then filter the “ticker2025” with the number 253010 to obtain the result.

The list of stocks are: 'ABNB','BKNG','CCL', 'CMG', 'CZR', 'DPZ', 'DRI', 'EXPE', 'HLT', 'LVS', 'MAR','MGM', 'NCLH', 'RCL', 'SBUX', 'WYNN', 'YUM'.

	ticker	GISC code	GISC sectors	GISC industries	mkt_cap_2008	mkt_cap_2025
133	ABNB	25301020	25	253010	NaN	84360.6532
227	BKNG	25301020	25	253010	4417.8866	163010.9096
261	CCL	25301020	25	253010	34253.8487	32154.0606
290	CMG	25301040	25	253010	4311.9660	81605.6948
335	CZR	25301010	25	253010	NaN	6924.7353
366	DPZ	25301040	25	253010	793.1789	15018.9932
368	DRI	25301040	25	253010	3781.4856	21849.0471
409	EXPE	25301020	25	253010	8691.5084	23786.4513
494	HLT	25301020	25	253010	NaN	59755.3060
596	LVS	25301010	25	253010	35628.5066	36178.8063
605	MAR	25301020	25	253010	12147.2149	76267.8519
609	MCD	25301040	25	253010	68718.2147	209618.4244
622	MGM	25301010	25	253010	24429.4084	10021.9446
663	NCLH	25301020	25	253010	NaN	11397.2386
760	RCL	25301020	25	253010	8770.0380	61577.8075
787	SBUX	25301040	25	253010	14104.0245	104502.3460
949	WYNN	25301010	25	253010	12504.2890	9202.4947
959	YUM	25301040	25	253010	19266.0595	37272.8368

Q3:

- 1) Filter the adjusted price data with the list of stocks
- 2) Calculate log returns for those stocks from 2021/01/04 to 2025/12/31.

```
log_ret = (adj_price_2125.set_index("Date").apply(lambda x: np.log(x) - np.log(x.shift(1))))
log_ret.reset_index(inplace=True)
log_ret = log_ret[(log_ret['Date'].dt.year>=2021) & (log_ret['Date'].dt.year<=2025)].copy()

log_ret.head()
```

- 3) Check and find that there is no missing value for log return

4) Obtain the simple average return of industry.

```
log_ret['industry simple average return'] = log_ret.mean(axis=1,numeric_only=True)
```

5) Regress return of MCD on simple average return of industry.

The results are as follows:

Alpha = 0.000307

Beta = 0.248171

R-squared = 0.14819

Date	ABNB	BKNG	CCL	CMG	CZR	DPZ	DRI	EXPE	HLT	LVS
2021-01-04	-0.053519	-0.028781	-0.060913	-0.049977	-0.036612	-0.010381	-0.024475	-0.007812	-0.034752	-0.028076
2021-01-05	0.063685	0.011024	0.010737	0.026926	0.043322	0.004837	0.015875	0.042986	0.004549	0.025387
2021-01-06	-0.038002	0.027826	-0.009756	-0.008672	0.019470	0.009216	0.023431	0.044777	0.025788	-0.015256
2021-01-07	0.057831	-0.004437	0.016529	0.018256	0.023592	-0.004375	0.027979	-0.007910	0.022495	-0.006685
2021-01-08	-0.009966	0.018442	-0.013104	0.024761	0.009435	0.014612	0.003222	0.012084	0.007650	-0.003963
...	...	...	...	...	...	...	...	...	...	...
2025-12-24	0.002122	0.001330	-0.013035	0.002907	0.002452	-0.001761	0.015441	-0.005776	0.006568	0.001660
2025-12-26	0.000292	-0.001170	-0.017757	-0.002377	0.011767	0.000306	-0.008142	0.003449	0.000614	-0.001660
2025-12-29	-0.001463	0.000219	0.000651	-0.017879	-0.031137	-0.010179	-0.007780	-0.002577	-0.003824	-0.008495
2025-12-30	0.002120	-0.002609	0.003250	-0.002696	-0.021452	0.001589	-0.010542	-0.006366	-0.006623	-0.008261
2025-12-31	-0.008730	-0.013322	-0.009127	-0.001080	-0.005543	-0.012161	-0.005041	-0.005947	-0.011010	-0.000154

MAR	MCD	MGM	NCLH	RCL	SBUX	WYNN	YUM	industry simple average return
-0.055723	-0.020525	-0.059156	-0.069190	-0.054619	-0.036942	-0.053988	-0.025563	-0.039500
0.010524	0.005976	0.021977	0.021264	0.025548	0.003292	0.030312	0.000094	0.020462
0.025835	-0.002276	0.026652	0.003706	-0.004835	0.007224	0.005973	0.005466	0.008142
0.007392	0.004638	-0.018777	0.013472	0.024354	-0.008094	-0.009973	-0.007642	0.008258
-0.005076	0.018183	0.024533	-0.009371	-0.013746	0.022199	-0.006583	0.014571	0.005994
...	...	...	...	...	...	...	...	...
0.003465	0.007979	0.003241	0.002593	0.006241	0.008431	0.000480	-0.001360	0.002388
0.001332	-0.008494	0.015782	-0.015659	-0.025745	0.006012	-0.005293	-0.007023	-0.002987
-0.003714	-0.006944	-0.013897	-0.017692	-0.010486	0.005743	-0.016949	-0.006350	-0.008486
-0.001496	-0.001622	-0.011639	0.000892	-0.003508	-0.003747	-0.015994	-0.000591	-0.004961
-0.011856	-0.007822	-0.006556	-0.004916	-0.009918	-0.012274	0.000000	-0.005932	-0.007299

Q4:

1) Obtain the market cap data for stocks in the list from year 2021 to year 2025.

2) Calculate the weights data

```
weights = mkt_cap_2125.div(mkt_cap_2125.sum(axis=1),axis=0)
weights
```

3) Multiply the log return with weights element-wise

4) Calculate the weighted average return of industry

```
df_ret_w = log_ret_stk*weights
df_ret_w['industry weighted average return'] = df_ret_w.sum(axis=1)
```

5) Regress return of MCD on weighted average return of industry.

Alpha = 0.0001746

Beta = 0.4230

R-squared = 0.2774

industry weighted average return
-0.035571
0.018017
0.003582
0.008686
0.010249
...
0.003690
-0.004196
-0.004606
-0.002816
-0.008983

The return of MCD are the same as Q3.